

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

How could activity distributed across cortex support one experienced situation? We separate three requirements: *coherence* (collective phase order), *compatibility* (local descriptions agree where their domains overlap) and *self-representation* (the state carries an accurate reconstruction of itself). Coherence supplies neither of the others, and five results establish that with no cortical hypothesis: synchrony constrains content only through encoders shared on overlaps, and uniformly over pairs only inside a patch too small to span the territories at issue; the resultant does not determine the state; a unique fixed point is not a self-model; where the encoding is a region's restriction, the region must resolve the relevant family before any readout is chosen; and agreement about a change takes communication. Three conditional results replace the uniform bound by a count of the pairs that can disagree, a chain across a cover's shared sites, or a spectral gap at a frequency-locked configuration, each against a declared input coherence does not supply. Four are checkable for a machine candidate, fixing what a claim of machine unity commits its maker to measuring. Under stated hypotheses, compatible local states glue

to a unique global state that an accurate contracting reconstruction fixes uniquely; installed energy bounds a declared actuator’s mean coupling, so a coherent branch requires it to exceed $2D/\kappa$, for a κ derived here for neither cortex nor a processor. A Lean 4 development carries these results and their composition, with the eight connecting hypotheses as explicit arguments, no incomplete proofs and an audited axiom footprint. One prediction needs no absolute calibration: onset delay scales as the square root of the rate at which coupling crosses threshold, which a within-subject emergence-from-anaesthesia design can test. We propose the glued state and its internal reconstruction as correlates of unity and a minimal self, and an extracellular-field hypothesis linking awakening to rising coupling; testing it requires calibrated geometry, resolved phase dynamics and validated decoders.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. The framework’s empir-

14 ical content is the coupling-gated onset prediction and its spatial signature.
15 The identification of the latter two relations as formal correlates of conscious
16 unity and a minimal self serves as interpretive motivation: it says why the
17 mathematics might matter, but does not constitute a testable proposal with-
18 out a proposed measurement on the experiential side.

19 Unity has been approached differently elsewhere: integrated information
20 theory treats integration and exclusion as constitutive (Tononi et al., 2016),
21 global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene
22 et al., 2014), and predictive-processing accounts derive both from a hierar-
23 chical generative model (Friston, 2010). Here the three requirements are
24 kept apart rather than derived from one another, so that each carries its own
25 assumptions and admits its own measurement; Section 10.3 returns to the
26 comparison.

27 Consider reaching for a cup. A visual description of its location and a
28 description of the hand’s relation to it can agree about where the cup is while
29 contributing different contents. Synchronizing the two processes would leave
30 disagreement about that shared quantity unresolved. An internal representa-
31 tion must additionally locate the perceiver within the represented situation.
32 This example motivates the requirements; the mathematical constructions
33 use spatial measures and fields in real normed vector spaces, whose identifi-
34 cation with perceptual content is a further modelling task.

35 The mathematical contribution is to specify connections between these
36 requirements and the assumptions each connection needs. A declared ac-

37 tuator’s stored energy bounds its mean coupling. Phase encoders shared on
 38 overlaps convert phase order into a bound on content disagreement. Compat-
 39 ible spatial measures have a unique global extension; approximate agreement
 40 controls a specified global selection. A contracting reconstruction makes that
 41 extension its unique fixed point only if it accurately reconstructs the exten-
 42 sion. Accuracy across a family of states imposes a further condition on the
 43 information retained by the encoding.

44 The empirical contribution is to identify what observations can decide.
 45 Stationary phase distributions can be shared by stronger coupling, weaker
 46 diffusion and a matched common drive. Finite ramps change the observable
 47 onset. A winding field is stationary at the same coupling as a uniform one and
 48 has the opposite resultant, so that observable does not determine the state.
 49 Coherent adaptation can lose alignment with a designated environmental
 50 template, and a compatibility estimator needs diagnostics that distinguish
 51 content from decoder artifacts. These findings determine the measurements
 52 needed to test the proposed connections.

53 Three of the ingredients are standard: the synchronization threshold
 54 $K_c = 2D$ for the noisy mean-field model (Sakaguchi, 1988; Strogatz &
 55 Mirollo, 1991), the unique gluing of compatible local sections, and the Ba-
 56 nach fixed-point theorem (Banach, 1922). Four things here are not. The
 57 installed-energy condition makes coupling a purchased quantity: a declared
 58 actuator’s stored energy bounds the mean coupling it supplies, so coherence
 59 on the stationary branch carries a resource requirement in units a measure-

60 ment can reach, and for cortex that requirement reduces to a figure the mode
61 prices must deliver. The coherence-to-content bound converts phase order
62 into a statement about descriptions rather than about timing, and its patch-
63 local form is what keeps it from being vacuous in a large population. A
64 stability criterion on the reweighted coupling kernel, together with an exis-
65 tence condition on an amplitude layer, settles when a low resultant reports
66 weak coupling and when it reports a winding field. And the composition is
67 machine-checked together with a witness discharging all eight hypotheses at
68 once, which is what makes their conjunction non-empty rather than a list no
69 object satisfies.

70 Granting those premises, collected in Table 1, the account runs as follows.
71 Extracellular contraction on awakening raises endogenous field amplitude
72 and with it the effective coupling between cortical oscillators. Once that
73 coupling passes $2D$, a coherent phase branch exists, and from an initial spread
74 within a quarter turn the population settles onto it; other initial data leave
75 windings stationary at the same coupling. Encoders shared where cortical
76 territories overlap turn that phase order into agreement about the quantities
77 the territories describe in common, so the local descriptions glue: there is
78 one state of which each is a restriction. Part of that state encodes the state
79 itself, and a readout of that part reconstructs it accurately enough, and
80 sensitively enough across the states that matter, to make it the unique state
81 the system models itself as being. On this account the glued state is what
82 makes distributed activity one situation, and its accurate self-reconstruction

83 is what gives that situation a perspective. Every step of it is conditional.

84 Section 2 collects the results that hold whatever physical field is proposed
85 and whatever substrate carries it; a reader who rejects the cortical hypothesis
86 of Section 9 still has to accommodate them. Sections 3–5 then develop the
87 three requirements together (Fig. 1), and Section 6 reads the unconditional
88 results against a machine candidate. Section 7 composes them, stating the
89 thermodynamic constraints and all eight connecting hypotheses per branch;
90 its supporting models of feedback, learning and resource supply have distinct
91 state spaces and assumptions. Section 8 then examines what the observables
92 can establish, before Section 9 develops the cortical hypothesis and an awak-
93 ening experiment. The hypothesis is that extracellular contraction increases
94 field-mediated coupling; its direction, rate and relation to recovery require
95 independent calibration. Supplemental Material organizes the constructions
96 and controls behind the claims they support. Table 2, at the end, lists the
97 symbols and the section that defines each.

98 **2 The landscape any phase-coherence ac-** 99 **count inherits**

100 Most of what follows is conditional. Table 1 names eight hypotheses that the
101 composition consumes rather than derives, and the cortical identification of
102 Section 9 is a further commitment on top of them. The five results below, by
103 contrast, are not conditional in that way. The mathematical features they

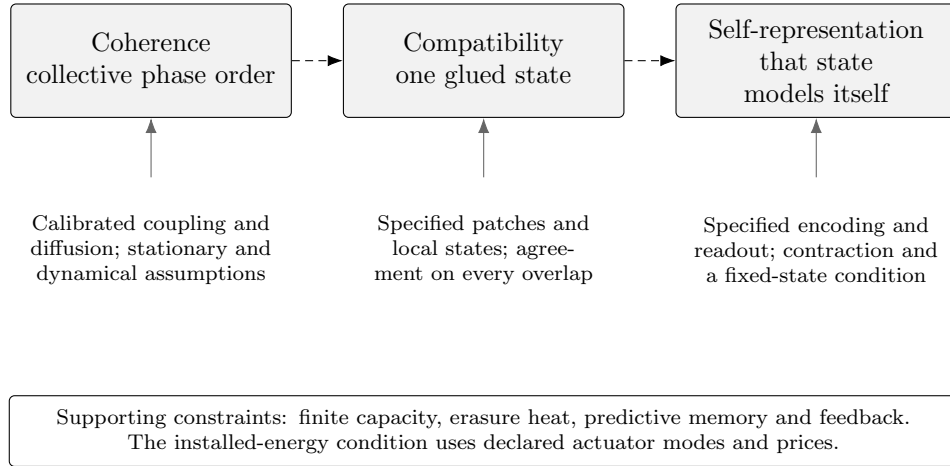


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

104 rely on reflect standard properties of phase oscillators (Acebrón et al., 2005;
 105 Ott & Antonsen, 2008; Dörfler & Bullo, 2014; Breakspear, 2017), and this
 106 framework’s contribution is tracking their implications for content decoding.
 107 They constrain any account that reads unity off phase coherence, whatever
 108 physical field is proposed to carry the coupling and whatever substrate carries
 109 the field. Each is stated below where it is used; collected here, they are the
 110 inherited constraints of the problem.

111 **Phase order constrains content only inside a patch too small to**
 112 **carry unity.** Assign every patch the same phase and let two local mass

113 profiles disagree at a shared site. The profiles are chosen independently of
 114 phase, so synchrony places no constraint on their disagreement, and they
 115 have no common extension: perfect locking is consistent with local descrip-
 116 tions that do not glue (Section 4). What the positive result of Section 4.2
 117 costs is exactly this gap. A bound on content disagreement needs encoders
 118 already related on overlaps, and that relation is a supplied fact about the de-
 119 scriptions rather than a consequence of the dynamics. Supplying it relocates
 120 the limit rather than lifting it. Eq. (6) bounds every pair at once and a chord
 121 never exceeds 2, so at the patch-local order of the sheet in Section 8.3 that
 122 uniform reading carries information only at 3 sites or fewer. The grain is the
 123 uniform reading's and not phase order's, and Section 4.2 states three results
 124 without it: a scale-free count of the pairs that can disagree, which names
 125 none of them; a bound chained through the sites a cover's patches share,
 126 which travels a distance linear in the hops taken; and, at a frequency-locked
 127 configuration, a bound read off the coupling graph's algebraic connectivity,
 128 which carries no population count at all. What none of the three changes is
 129 the constraint stated here: a count locates no site, and the other two con-
 130 sume a declared cover or declared hardware data, so phase order by itself
 131 still says nothing about territories too far apart to share a patch, which is
 132 where the compatibility claim is made.

133 **The order parameter does not determine the state.** A field wind-
 134 ing once across a periodic sheet has resultant $r = 0$ — the order parameter

135 of Eq. (1) below — while every patch of it stays locked, and this is not a
 136 contrived initial condition. On a periodic array of any dimension, a winding
 137 is stationary for *any* isotropic symmetric kernel at identical frequencies, be-
 138 cause the coupling term is odd in the separation against a kernel even in it
 139 (`char_is_kuramoto_trajectory`). Section 8.3 exhibits a sheet on which uni-
 140 form and winding initial conditions reach 0.9657 and 0.0083 at one coupling,
 141 diffusion and kernel. A low resultant therefore reports either weak coupling
 142 or a winding field, and no measurement in r alone separates them. The state
 143 also carries an integer the resultant does not: read around a ring, the same
 144 winding has winding number q , and a nonzero q obstructs every global real-
 145 valued phase lift (`winding_degree_obstructs`). The resultant is an average
 146 over sites and moves when they are reweighted or the patches redrawn; the
 147 degree telescopes and does not.

148 **A unique fixed point is not a self-model.** A constant readout con-
 149 tracts, fixes whatever state it returns, and reconstructs nothing. Uniqueness
 150 is consequently the wrong criterion, and Section 5.2 gives the right shape
 151 of one: reconstructing M states pairwise more than 2ε apart within error ε
 152 requires M distinguishable codes, so an alphabet of 2^b codes caps the family
 153 at $M \leq 2^b$. This is a counting argument and holds for any encoder and
 154 readout whatever.

155 **Where the encoding is a restriction, the region must resolve the**
 156 **family.** Read the code as the state's own restriction to a region and

157 the criterion moves off the readout entirely. A boundary whose region
 158 reads two relevant states identically cannot reconstruct them more than 2ε
 159 apart (`not_isRestrictionResonance_of_reconstructs`), and a resonant
 160 boundary that does reconstruct a separated family restricts injectively on
 161 it (`restrictToAvatar_injOn_of_isRestrictionResonance`). Where the
 162 region sits is fixed before an encoding is chosen, so no readout supplies
 163 resolution the region lacks.

164 **Agreement after a change takes communication.** In a synchronous
 165 network whose updates read only the site’s own state and its declared neigh-
 166 bours’ messages, two configurations agreeing on a site’s causal past through
 167 T rounds leave that site identical at round T . Two interventions outside
 168 that past that require reports more than 2ε apart therefore cannot both be
 169 reported from it within ε by the deadline (Section 5). Snapshot compatibil-
 170 ity and timely compatibility are therefore different requirements, and cutting
 171 the path removes the second while leaving the first intact.

172 None of these depends on the coupling being electromagnetic, on the sub-
 173 strate being cortex, or on the eight hypotheses of Table 1 holding. Section 6
 174 applies four of them — all but the one about the resultant, which is a state-
 175 ment about a spatial observable — to a candidate that is neither biological
 176 nor field-coupled.

177 **3 Coherence: physical coupling and its re-** 178 **source condition**

179 **3.1 Dynamics and the stationary threshold**

180 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

181 Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase dif-
182 fusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units,
183 with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; fre-
184 quency heterogeneity and diffusion are distinct parameters. The stationary
185 calculation below uses identical frequencies and this scalar coupling. Ap-
186 plying it to a spatial kernel requires an aggregation rule and a check of the
187 resulting approximation.

188 For identical frequencies in their rotating frame, the positive normalized
189 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

190 where I_n are modified Bessel functions and ψ is the collective phase. The for-
191 mal derivation uses the absence of stationary current for a periodic gradient
192 drift to obtain this density, then verifies normalization and self-consistency.

193 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988; Strogatz &
 194 Mirollo, 1991). Above threshold, exactly one positive solution coexists with
 195 zero and increases strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

196 Figure 2 shows this stationary branch.

197 Linearization about the uniform density gives rate $K/2 - D$ for the first
 198 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
 199 exactly above threshold. About the coherent density, the quadratic form
 200 is strictly dissipative transverse to collective rotation, whose rate is zero.
 201 These linear estimates concern the limiting equation. Relating them to finite
 202 stochastic trajectories requires a dynamical mean-field limit and branch selec-
 203 tion; exchange symmetry alone does not establish the needed independence
 204 (Sznitman, 1991).

205 **3.2 Installed energy bounds mean coupling**

206 A spatial model replaces the finite sum by integration against a kernel. One
 207 declared implementation carries response modes with continuous profiles ϕ_i ,
 208 installation prices u_i and nonnegative occupancies $c_i(\Xi)$ in configuration Ξ :

$$K_\Xi(x, y) = \sum_i c_i(\Xi) \phi_i(x) \phi_i(y), \quad U(\Xi) = \sum_i u_i c_i(\Xi). \quad (4)$$

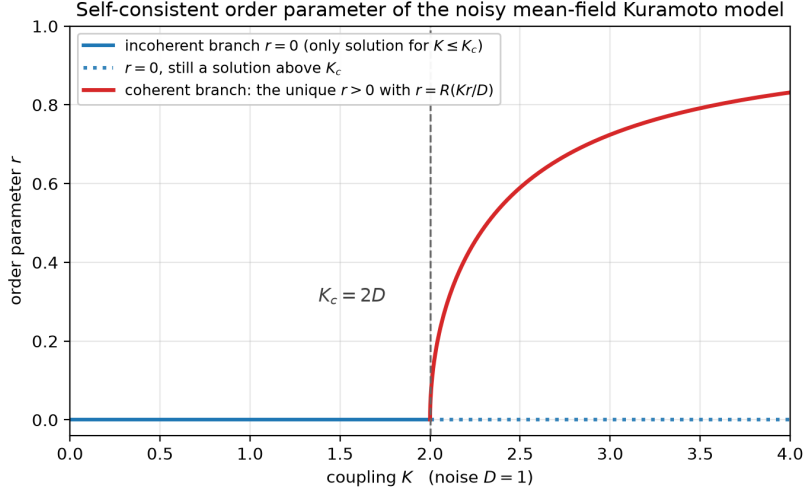


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

209 The same configuration sets the reciprocal kernel and stored energy. An
 210 executed transition changes the kernel through its changed occupancies and
 211 requires work $U(\Xi') - U(\Xi) + Q$, with Q positive into the reservoir. The
 212 installed kernel is averaged over the resulting configuration law $p(\Xi)$. Cell-
 213 pair reconstruction on refining spatial partitions converges to its product-
 214 measure integral under the specified compactness and sampling assumptions;
 215 the refinement index is independent of the number of control updates.

216 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 217 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's

218 integral gives

$$\begin{aligned}
U_{\text{inst}} &= \sum_{\Xi} p(\Xi) U(\Xi), \\
K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_{\Xi} p(\Xi) \sum_i c_i(\Xi) m_i^2 \leq \kappa U_{\text{inst}}.
\end{aligned} \tag{5}$$

219 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
220 coherent stationary branch and any growing Fourier mode about the uniform
221 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
222 is necessary: energy above it need not produce sufficient coupling because
223 the bound can be loose.

224 The quantity constrained is installed energy rather than dissipated heat,
225 and the condition binds through κ rather than through the energy, since
226 any macroscopic substrate carries installed energy far above the thermal
227 floor. Read forwards it therefore excludes no macroscopic candidate. Read
228 backwards it is a number. Take the region the field arithmetic takes, a
229 sphere of one LFP decay length holding 1676 neurons, at the metabolic rates
230 Supplemental Material collects and over one phase-diffusion time $1/D = 2$ s
231 at $D = 0.5 \text{ rad}^2/\text{s}$: the installed energy is $1.0 \times 10^{-6} \text{ J}$, and the condition is
232 met at every κ above $9.6 \times 10^5 \text{ J}^{-1}\text{s}^{-1}$. That figure is what an identification
233 of cortical mode prices has to deliver, and the first identification delivering
234 less refutes the coupling-gated account for cortex at this region and interval.
235 Another interval rescales the figure by its ratio to this one.

236 What the condition discriminates, it discriminates through that factor.

237 The required κ scales as D^2 and inversely with the region's cell count, so at
 238 fixed mode prices the condition ranks candidate substrates by their installed
 239 energy. In a declared implementation it decides outright, the learning witness
 240 of Section 7 clearing the line after one update while its matched control stays
 241 on the exclusion side at the same conversion factor and diffusion. What is
 242 missing for cortex is a price list rather than a mechanism. A mode is a priced
 243 rank-one contribution to the kernel, cortical field generation has no canonical
 244 decomposition into such modes, and choosing the basis would choose the
 245 answer. Applying the scalar threshold to heterogeneous or signed spatial
 246 dynamics requires a justified reduction, and identifying the installed kernel
 247 with a cortical field is a separate physical commitment.

248 **4 Compatibility: from coordinated activity** 249 **to a global description**

250 **4.1 Why common timing is insufficient**

251 For overlapping patches U and V , compatibility means that restricting either
 252 local description to $U \cap V$ gives the same object. Descriptions can agree on
 253 shared quantities while contributing different information elsewhere. Com-
 254 patible sections of the specified sheaf have a unique global extension. On
 255 a finite cover, local finite spatial measures agreeing as measures on every
 256 overlap also have a unique finite-measure extension whose sheaf image is the

257 glued section. No mass-one normalization is imposed.

258 In the cup example, visual and reaching descriptions must express a
259 shared quantity, such as cup location, in common coordinates. Restriction
260 retains that quantity while dropping details particular to each description.
261 Agreement permits surface appearance and hand configuration to contribute
262 to one situation. The mathematical counterexample assigns the same phase
263 to every patch while two local mass profiles disagree on a shared site. Those
264 profiles have no common extension. Because they are chosen independently
265 of phase, synchrony places no constraint on their disagreement.

266 This distinction limits the biological interpretation. Receptive territories
267 and content domains must be related to spatial patches, and the descriptions
268 must support the required extension. Matching overlap marginals of de-
269 coded probability laws need not determine a unique global joint distribution
270 (Abramsky & Brandenburger, 2011).

271 The cover is itself a choice, and the construction constrains it in one way
272 only: synchronized patches agree where they overlap. Where they do not
273 overlap there is nothing to compare, so a cover by pairwise disjoint patches
274 satisfies that obligation for *every* assignment of local descriptions and phases
275 (`ofDisjointCover`) and the gluing theorem still returns a unique global sec-
276 tion, which is the tuple it was handed (`glue_unique_of_disjoint`). Member-
277 ship of the class then reports nothing about the system beyond the patches
278 covering it. What a gluing claim asks of a cover is therefore a non-degeneracy
279 the theorem does not assume: that distinct patches meet (`HasNonempty-`

280 Overlaps), and that what they share is a quantity fixed independently of
 281 the agreement it is being used to establish.

282 4.2 When coherence does constrain content

283 A declared relation between phase and content gives a positive result. Let
 284 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 285 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 286 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 287 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 288 sites; their equality on overlaps is a supplied relation between descriptions of
 289 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 290 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every
 291 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (6)$$

292 Here discrepancy is the norm of the difference, and r and N are read on
 293 whichever population the bound is applied to. That choice is what decides
 294 whether the estimate says anything. A patch carries an order parameter of
 295 its own, and reading the bound there carries that patch's resultant and that
 296 patch's site count (`chord_le_of_patch_coherence`, `compatible_of_patch_`
 297 `coherence`), so a large population observed through small patches is con-
 298 strained by the small count. Read instead on the whole array, the factor N

299 makes the estimate weak, because it draws a pointwise guarantee from an
300 average over N^2 pairs. At $r = 1$ the observations agree exactly at either
301 scale. A shared scalar encoder is a special case. Dropping the overlap rela-
302 tion permits disagreement even at perfect locking. A common nonexpansive
303 linear projection preserves agreement, but agreement of a projection does
304 not establish agreement of the full vector.

305 What Eq. (6) bounds is a chord, which never exceeds 2, so the estimate
306 constrains anything only where the right-hand side falls below that. At
307 the sheet’s own patch order, the bound is informative for at most 3 sites;
308 beyond that, the coherence-to-content link requires the bridge assumption
309 (E78) of the cortical hypothesis. Supplemental Material provides the detailed
310 derivation of this saturation, and states three conditional derivations that
311 replace the uniform factor N by a count of disagreeing pairs, a chain across
312 a cover’s shared sites, or a spectral gap at a frequency-locked configuration.

313 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
314 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
315 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
316 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

317 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
318 duce an initial mismatch. Approximate observations leave a nondecaying
319 term in the error bound. Taking a uniform bound on Eq. (6) connects this

320 update to the phase process.

321 4.3 What approximate agreement determines

322 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
323 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
324 Weights that vanish outside their patches and sum to one at each site select
325 a global profile by weighted averaging. The selection lies within ε of each
326 patch there, and any two such selections differ by at most ε , independently
327 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
328 $L\varepsilon$ across selections.

329 At zero tolerance, the selection is the unique exact extension. At every
330 positive tolerance there are families with no exact extension; selection then
331 gives an approximation whose dependence on weights is bounded. Supple-
332 mental Material gives a three-site witness attaining the bound. The mass-
333 profile construction uses the scalar case of the content dynamics and addi-
334 tionally requires nonnegativity of the initial profiles and observations. The
335 general vector result supplies no identification with the spatial-measure sheaf.

336 These constructions concern declared objects: a specified cover, contents
337 valued in a normed space, and encoders related on overlaps. They select
338 neither the neural cover nor the decoder, and identifying their contents with
339 perceptual content, or establishing that the identification bears on experi-
340 ence, is a further modelling task; Section 8 sets out what a measurement of
341 it would have to establish.

342 5 Self-representation: a state that models it- 343 self

344 5.1 The reflexive construction

345 Compatibility gives a global state. Reflexivity relates it to an encoding within
346 that state. Equip global sections with a complete metric. If the additional
347 restriction-resonance condition identifies encoding with restriction to a region
348 $\mathcal{A}$, the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (8)$$

349 The region and readout are fixed across the state space. If F contracts and
350 fixes the section s_* formed from the local descriptions, Banach’s theorem
351 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89 sup-
352 plies the contraction and $F(s_*) = s_*$; restriction resonance is separate. Su-
353 percritical coupling makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$
354 less than one for a relaxation time $\tau > 0$. Constructing a map with that
355 factor remains a modelling obligation.

356 In the cup example, the global state includes the represented scene and the
357 perceiver’s relation to it. A local encoding must retain enough information
358 for the readout to recover that modelled state; cup position alone would be
359 insufficient if hand configuration and viewpoint could not be reconstructed.

360 For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruc-

tion gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (9)$$

At the stipulated rate this radius grows as $2\varepsilon/((K - 2D)\tau)$ near threshold, for fixed positive ε and τ . Exact uniqueness survives while the error guarantee weakens.

5.2 Reconstruction across a family of states

Sensitivity can be required quantitatively. Declare relevant macrostates, an encoder Φ and readout R , and require $d(s, R(\Phi(s))) \leq \varepsilon$ throughout that family. If two states have the same code, the triangle inequality gives

$$\Phi(s) = \Phi(t) \implies \max\{d(s, R(\Phi(s))), d(t, R(\Phi(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

Thus M relevant states pairwise more than 2ε apart require at least M distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$. Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the existing self-prediction map.

This is an additional accuracy criterion on a family, with no requirement for multiple exact fixed points. A blind encoder or constant readout fails it for separated states despite having a unique fixed point.

Where the encoding is read as the state's own restriction to a region, the criterion becomes a condition on the region. Restriction resonance requires the encoder and that restriction to identify the same pairs of states, so a

379 boundary reconstructing two relevant states more than 2ε apart while its re-
 380 gion reads them identically is not resonant (`not_isRestrictionResonance_`
 381 `of_reconstructs`). Read forwards, a resonant boundary that reconstructs
 382 a pairwise separated family restricts injectively on it (`restrictToAvatar_`
 383 `injOn_of_isRestrictionResonance`). Which states a region reads identi-
 384 cally is fixed by where the region sits, before an encoding is chosen, so the
 385 resolution the family demands is the region's and no readout supplies it. The
 386 implication runs one way: a region that does resolve the family is not thereby
 387 shown to carry a resonant encoding, and E89 supplies no such resolution.

388 Those two demands are jointly satisfiable with nothing folded anywhere.
 389 Take the region to be the whole substrate: restriction is then the identity, so
 390 the boundary is resonant, reconstructs every declarable family exactly, and
 391 restricts injectively on all of them (`topAvatar_satisfies_obligations`). A
 392 properness condition on the region (`IsProperAvatar`) is therefore a separate
 393 requirement rather than a consequence of resonance and resolution, and what
 394 a fold into a local region is asked for. Its quantitative form replaces the
 395 alphabet by a metric. With an L -Lipschitz readout, a family reconstructed
 396 within ε whose members lie pairwise more than $d_{\min}$ apart has δ -separated
 397 codes whenever $\delta L \leq d_{\min} - 2\varepsilon$, so the family is no larger than the δ -packing
 398 number of what the region carries (`encard_le_packingNumber_restrict-`
 399 `ToAvatar`). That is the shape a continuous family of states needs, and at
 400 the whole substrate it says nothing, because the region's range is everything.
 401 The gain L is declared and derived here for no physical readout, and at

402 unbounded gain the bound says nothing.

403 **5.3 Agreement after a change takes communication**

404 Snapshot compatibility does not specify how quickly descriptions respond
405 when a shared quantity changes. In a synchronous network whose updates
406 read only the site’s own state and declared neighbours’ messages, two initial
407 configurations agreeing on a site’s causal past through T rounds produce the
408 same state there at round T . The deadline consequence is the one Section 2
409 states, and it follows from the reconstruction bound in Section 5.2: a report
410 from that site is a readout of a code that the two interventions leave identical.

411 A three-site delay line witnesses failure after one round and exact report-
412 ing after two; cutting the path prevents delivery at every round. A report
413 may still happen to be correct about one fixed world. The bound concerns
414 responsiveness across interventions.

415 Throughout this section *prediction* names a reconstruction map, and the
416 relevant family of states, the metric, the tolerance, the communication graph
417 and the deadline are all declared inputs, as are the represented variables and
418 the fidelity counted as accurate. Reading the map as temporal neural pre-
419 diction would require dynamics; reading the resulting code count as physical
420 memory capacity would require a hardware model, which a state-dependent
421 readout would have to extend.

422 6 What the conditions constrain in a digital 423 candidate

424 The results of Section 2 hold for any candidate, and for a large language
425 model (LLM) running on graphics processing units (GPUs) four of them are
426 checkable rather than merely stateable. Those four are where this framework
427 bears on claims made for such systems. It does not exclude them, and no re-
428 sult here could; what it does is fix what a claim of unity or self-representation
429 commits its maker to measuring.

430 **A quantitative bound for a linear bottleneck.** Let a finite spatial
431 coupling operator K have orthonormal eigenmodes e_i with $|\lambda_1| \geq \dots \geq |\lambda_N|$.
432 Every linear approximation A of rank at most d satisfies

$$\sum_{i=1}^N \|Ke_i - Ae_i\|^2 \geq \sum_{i=d+1}^N \lambda_i^2. \quad (11)$$

433 Retaining the first d modes attains equality. A linear encoder and readout
434 factoring through d real coordinates therefore pay this error. Applying the
435 bound requires a spectrum for the target and a rank constraint on the actual
436 approximation. Query/key dimension alone supplies no such constraint on
437 softmax attention: even a two-position rank-one logit matrix can yield rank-
438 two attention. The result prices an explicitly linear bottleneck in a finite
439 representation of a spatial kernel; it gives no universal gap between trans-

440 formers and continuous fields. Supplemental Material derives the bound and
441 checks the softmax counterexample.

442 **Codes, not reports.** Reconstruction across a family is a counting con-
443 straint before it is anything else. If two macrostates the candidate is said
444 to represent satisfy $\Phi(s) = \Phi(t)$ while $d(s, t) > 2\varepsilon$, Eq. (10) rules out re-
445 constructing both within ε , whatever the readout. A system’s reports are
446 readouts. Identical verbal reports across two conditions are therefore consis-
447 tent with identical internal codes and settle nothing about the states. The
448 converse claim—that the system represents its own condition across a de-
449 clared family—asserts that its internal code separates that family, which is a
450 statement about activations under intervention and not about text. Naming
451 the channel makes the count concrete. Where the code is the token sequence
452 a model emits—which is what its next step reads—the alphabet is the de-
453 ployed vocabulary (`tokenChannel`), and a family reconstructed through it
454 is no larger than the vocabulary size raised to the number of steps (`card_`
455 `le_card_tokens`). Over an extended exchange that number is large and the
456 bound slack. At a single step it is one vocabulary’s worth, so claiming that a
457 single emission reports the model’s own condition across a declared family as-
458 serts that the vocabulary separates it (`card_le_card_tokens_one`). A claim
459 resting on activations rather than on text has declared a different channel,
460 to which the same count applies with that channel’s alphabet. The measure-
461 ment follows from the statement: declare the family and the fidelity first,

462 then test whether the encoding region distinguishes its members.

463 The same interface has an information ceiling. For any finite joint law
464 of an internal state X and a length- k token sequence Y over vocabulary V ,
465 $I(X; Y) \leq H(Y) \leq k \log_2 |V|$ bits. The proof uses the full joint law and
466 allows dependent emissions and zero-probability outcomes. A binary copy
467 attains one bit per emitted binary symbol; an independent output conveys
468 zero. This counts the declared output sequence, so a cache, timing signal or
469 other accessible state must be included separately when it forms part of the
470 channel.

471 **The region, not the readout.** Where the encoding is read as the state’s
472 restriction to a region of the machine, the demand falls on the region before
473 any readout exists. For a decoder-only transformer the causal mask makes
474 the network an instance of the update model of Section 5: the activation
475 at position t and layer ℓ reads positions at most t at layer $\ell - 1$ and noth-
476 ing else. A region’s causal past is therefore fixed by the architecture rather
477 than chosen by interpretation. That instance is formal, and sharper than
478 slow growth: the causal past of position t stays inside the positions at most
479 t at every depth (`mask_ball_subset_1e`). The deadline bound of Section 5
480 therefore fires at every deadline at once rather than at a chosen one, its com-
481 munication graph given rather than assumed (`mask_no_guarantee`). That is
482 a statement about one forward pass; across steps a model does read its own
483 prior output, which is the channel the preceding paragraph counts. Read

484 against the coupling proposed here the asymmetry runs both ways, and nei-
 485 ther way is an advantage. A kernel of full support has no neighbourhood
 486 structure for the bound to bite on: after one round the causal past is the
 487 whole substrate, so the hypothesis that fires the obstruction is unsatisfiable
 488 at any deadline but zero (`not_outside_past_of_isFullSupport`). A physi-
 489 cal field propagates at a finite speed, so treating one that way is a calibration
 490 against the phase time scale and not a theorem. What the comparison does
 491 fix is where the bound has force. Not on the attention graph, where a posi-
 492 tion reads every position at or before it and the fan-in is the context length.
 493 On the physical interconnect, fan-in is genuinely bounded: a graph whose
 494 sites each hear from at most ν others reaches at most $(T + 1)\nu^T$ sites in
 495 T rounds (`card_ball_le_mul_pow`), so a model sharded across many devices
 496 pays a number of rounds growing like $\log_\nu$ of the device count. Buying that
 497 deadline back costs wiring: reaching every site in one round requires a device
 498 hearing from all but one of the others (`le_degree_of_reaches_one`). Both
 499 ν and the device count are declared, as the communication graph and the
 500 deadline already are, and what the count bounds is the graph a guarantee
 501 requires rather than what hardware is buildable. A claim that a model re-
 502 ports on its own processing consequently carries an obligation to say which
 503 positions and layers the report's causal past contains. A report that appears
 504 to track an intervention outside it indicates a confound or a different region,
 505 not a faster channel.

506 **Agreement about what.** Unity requires the candidate’s local descriptions
 507 to agree on independently identified shared quantities (Section 4). Common
 508 timing, a shared residual stream and fluent output supply none of that, for
 509 the reason Section 2 gives: coordination bounds content disagreement only
 510 through encoders already related on overlaps. Which quantities two parts
 511 of a model describe in common is a decoding question, carrying the failure
 512 modes of Section 8. The cover is a choice, and choosing it to secure agreement
 513 empties the claim.

514 **What is not excluded.** Digital representations can satisfy compatibility,
 515 so finite hardware is compatible with the gluing construction and none of
 516 the above is a no-go for machines. The coupling-gated route E56–E78 is a
 517 different matter: it requires a calibrated physical coupling, its noise regime
 518 and a connection from coordination to content agreement, and an attention
 519 matrix or a shared clock supplies none of these identifications. Computing a
 520 field model likewise leaves its realization in the computing substrate to be es-
 521 tablished. Those physical obligations fall on the cortical proposal identically.
 522 The capacity comparison those support comparisons lack is available, and it
 523 favours neither side by construction. A continuum code space read at reso-
 524 lution δ holds at least its volume divided by the volume of a resolution cell,
 525 so it out-resolves an alphabet of 2^b codes exactly when that volume exceeds
 526 2^b cells (`two_pow_lt_packingNumber_of_lt_measure`). Below that threshold
 527 the inequality runs the other way, and at a coarse enough resolution the re-

528 gion holds a single code. What decides the comparison is the volume, the
529 resolution and the dimension, all measured quantities, and continuity decides
530 nothing on its own: a continuum code space read at a finite resolution is a fi-
531 nite alphabet whose capacity is a bit count. For the same reason this account
532 credits an analog medium with no computation beyond a Turing machine's.
533 Real-valued states carry unbounded capacity only at unbounded precision,
534 and the noise floor $D > 0$ that Eq. (1) is stated against removes it.

535 The installed-energy condition of Section 3.2 is restricted in the same
536 way. It constrains a macroscopic candidate only through κ , which is hard-
537 ware data supplied by mode profiles and prices, and Supplemental Material
538 reads the condition against a cortical metabolic budget to return the conver-
539 sion factor it would require rather than a verdict. The hardware comparisons
540 there have similarly restricted scope: fixed wiring can impose a correlation
541 disadvantage, and point-supported kernels vanish under an atomless contin-
542 uum measure, but neither comparison makes the representation conditions
543 unattainable on GPUs. A universal exclusion would need a necessary condi-
544 tion for consciousness and a proof that every relevant implementation violates
545 it, and the present composition supplies neither.

546 Whether experience requires a living substrate is the explicit subject of
547 biological naturalism (Seth, 2026a,b). On that view the properties ground-
548 ing experience are those of self-sustaining organisms, and simulating them
549 does not secure them. The conditions stated here name no biological mech-
550 anism, but computation described as such does not satisfy them either: a

551 candidate must exhibit a physically realized coupling in a measured noise
552 regime, agreement about independently identified shared quantities, and an
553 internal reconstruction meeting a declared fidelity. Whether a realization
554 of those conditions that is not alive would be accompanied by experience
555 is the point at issue, and these results do not settle it. A related position
556 locates the difference in the mode of computation rather than in life as such:
557 biological systems compute across scales that do not separate, and along-
558 side discrete operations carry continuous-valued dynamics belonging to their
559 physical medium (Milinkovic & Aru, 2026). The coupling condition stated
560 here is one measurable instance of that requirement, fixed to a physical field
561 in a declared noise regime, and a description of such a field does not satisfy
562 it.

563 The neuroscientific case against *human-like* consciousness in text-only
564 LLMs concerns embodied sensory interaction, thalamocortical organization
565 and biological regulation linking perception and action to survival (Aru et al.,
566 2023). Their absence supports skepticism, conditional on their causal rele-
567 vance to experience. Those mechanisms are not requirements of the abstract
568 gluing and reconstruction definitions, and workspace-like representations re-
569 ported in some LLMs (Gurnee et al., 2026) supply candidate organization to
570 investigate. What this framework adds to that investigation is an order for
571 the questions: declare the family of states and the fidelity, fix the encoding
572 region and read its causal past off the architecture, and only then ask what
573 the system reconstructs. A claim that survives those three steps is a claim

about the machine; one that does not survive them was a claim about its
output.

7 Thermodynamics and composition

The phase and representation results concern coupling, content disagreement and reconstruction error. Thermodynamics constrains the physical operations that could implement them, and it does so from the side rather than along the spine: no edge below derives coherence, compatibility or reconstruction from a heat bound, and the resource condition these edges terminate in is necessary without being restrictive for any macroscopic candidate (Section 3.2). What this half contributes is a ledger in which a resource claim is stated in checkable units. Finite-register capacity obeys $H(p) \leq \log |X|$; erasure heat depends on specified system–bath assumptions (Landauer, 1961; Norton, 2005; Shenker, 2000). A separate boundary result concerns a field reaching disconnected vacuum components (Kibble, 1976). The composition connects these objects through the explicit hypotheses in Table 1.

7.1 Probability current supplies a dissipation bound

For a positive normalized phase density ρ , diffusion $D > 0$ and current $J = v\rho - D\partial_\theta\rho$, define $\sigma_J = \int_{-\pi}^{\pi} J^2/(D\rho) d\theta$. Continuous density and current give the sharp bound

$$\left(\int_{-\pi}^{\pi} J d\theta\right)^2 \leq D\sigma_J. \quad (12)$$

593 A uniform density under constant drive ω attains $\sigma_J = \omega^2/D$. In contrast,
 594 the identical-frequency gradient model has positive self-consistent stationary
 595 order above $K = 2D$ with $J = 0$ and $\sigma_J = 0$. Thus neither positive order nor
 596 its time average forces positive dissipation: a stationary ordered density is
 597 already a counterexample. A cortical maintenance-power bound requires an
 598 identified nonequilibrium drive, its probability current and a thermal con-
 599 version; phase order alone does not supply them. The current functional
 600 is distinct from the substrate squared-drift objective used in the plasticity
 601 controls. It also bounds the rate of the cosine moment $m = \int \cos \theta \rho(\theta) d\theta$:
 602 $\dot{m}^2 \leq D\sigma_J$, when the classical continuity equation and sufficient regular-
 603 ity justify differentiating this moment. This constrains density change. It
 604 does not bound an externally chosen $\dot{K}$: the uniform density is an exact time-
 605 dependent solution of zero current cost for every coupling schedule, including
 606 positive schedules with any prescribed derivative.

607 7.2 Feedback changes the information constraint

608 For passive memory X_t driven by an environmental signal S_t , the predictive
 609 bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (13)$$

610 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
 611 et al., 2012). Zero retained information also satisfies it, so a task and a
 612 learning rule require additional specification. Relating this bound to Lan-

613 dauer heat requires a common thermal scale and budget, and memory waste
 614 bounded by erased entropy.

615 An acting agent changes the signal it will receive. Specify policy, world,
 616 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
 I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
 &\quad - I(X_t; S_{t+1} \mid S_t).
 \end{aligned}
 \tag{14}$$

617 The second conditional term vanishes in the passive model. With feedback,
 618 the signed difference can be negative as action creates correlations with the
 619 future. A causal reading requires an environmental state sufficient to describe
 620 the relevant dynamics.

621 For a controlled substep, the active thermodynamic bound instead uses
 622 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
 623 $\Delta H(X, S) + \langle Q \rangle / (k_B T) \geq 0$ under local detailed balance, with thermal scale
 624 $k_B T > 0$ and heat positive into the reservoir (Horowitz and Esposito, 2014;
 625 Ito and Sagawa, 2013). Every forward path of positive probability must have
 626 a positive-probability reverse for this finite identity. A missing reverse gives
 627 infinite divergence. The supplement develops the support conditions and
 628 composes actuation with memory updates on their shared intermediate law.

629 **7.3 Learning and repeated operation need explicit re-** 630 **sources**

631 The finite examples distinguish selecting a feasible policy, relaxing a register
632 toward an encoded reward, and learning an unknown task from observa-
633 tions. The observational learner acquires correlation with an environmental
634 parameter while its action marginal stays uniform. A continuing model exe-
635 cutes actuation, observation-driven learning and reset in sequence, preserving
636 above-chance performance at a positive work cost per cycle.

637 A phase-derived sensor makes the informational distinction concrete: an
638 informative readout and a blind control have the same phase order but dif-
639 ferent learning performance. Attaching the learner’s register to the actuator
640 of Section 3.2 ties reward, installed kernel and work to its own executed law.
641 The learning witness rises above the necessary installed-energy bound while
642 its matched control remains on the exclusion line. These finite constructions,
643 detailed in Supplemental Material, leave the evolving feedback between phase
644 dynamics, installation and sensing to be specified.

645 For that model, common energy observable $\mathcal{E}$, thermal scale $k_B T$, local
646 detailed balance and positive support at each stage give

$$k_B T [H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N \leq B + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N, \quad (15)$$

647 where expected cumulative work stays within the supplied allowance B at
648 every prefix. This bounds expectations. Separate constructions track a store

on each trajectory, include finite sources, charge sensor clearing and prepend preparation. They specify when a mean allowance represents an available resource; an expected budget alone permits overdrawn paths.

These constructions establish costs for their declared processes. A finite source with a positive net cost permits only finitely many funded cycles, while continuing operation requires replenishment. Register–bath witnesses also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 3.2 requires physical relations that these separate models do not supply automatically.

7.4 What the composition establishes

The passive theorem `chain` and feedback theorem `chain_active` each accept eight connecting implications (Table 1). In the active branch, a named substep at the register’s thermal scale satisfies

$$k_B T > 0, \quad k_B T [H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (16)$$

with B the register’s heat. The upper allocation follows from a ledger whose operations deliver that total heat to one reservoir and whose other operations do not increase joint entropy. Those ledger assumptions are physical inputs. Neither Landauer’s lower bound nor microscopic reversibility supplies them. The regime in which those premises are exhibited is a near-equilibrium one:

668 strictly positive forward and reverse elementary paths, local detailed balance
 669 at one thermal scale, and entropy differences and heats of order $k_B T$, the
 670 witnesses being two-bit systems with interaction energies of that size. Cortex
 671 is not in it. The installed energy of Section 3.2 is 2.4×10^{14} times kT at body
 672 temperature over one region and one phase-diffusion time, and a budget B
 673 of that size stands against an entropy term of order $k_B T$, so Eq. (16) is
 674 satisfied by any macroscopic candidate and excludes none of them. Reading
 675 the register instead at the scale where its premises are physical, such as a
 676 single channel, relocates the difficulty to E45Active, which would then have to
 677 carry one molecular step’s budget to the convergence of a continuum kernel.
 678 The active branch is an alternative route to the same node rather than a
 679 stronger one, and no result of Section 3.2 passes through it: the installed-
 680 energy condition is derived from E56 and E67.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12 <i>capacity and boundary</i>	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23 <i>boundary to register</i>	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.

Table 1 (continued)

Connection	Status	Required assumption
E34 / E34Active <i>register to dissipation</i>	Bridge assumption	Passive: predictive memory at the register's thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations' compressiveness. Both are exhibited at thermal-scale heats on finite state spaces with positive reverse paths, which is not the regime a cortical register occupies.
E45 / E45Active <i>couplings to continuum</i>	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Convergence follows from mesh refinement rather than from the budget, which selects no regime at macroscopic scale. Refinement and physical identifications remain inputs.
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.

Table 1 (continued)

Connection	Status	Required assumption
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover’s nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve’s diameter; where the configuration is frequency-locked it carries through the coupling graph’s algebraic connectivity λ_2 instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of κ .
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 3.2’s occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

681 Both branches also take a thermodynamic cover carrying exact over-
682 lap compatibility and a reflexive boundary specifying encoding and readout.
683 They conclude **UnifiedSelf**: the glued section is the unique fixed point of

self-prediction. Compatibility and accuracy at that section enter as premises;
uniqueness follows. Identifying encoding with restriction needs the separate
resonance condition of Eq. (8).

Each of the eight edges is supplied rather than derived from the one above
it. E12 is equivalent to its conclusion, so capacity and boundary are inde-
pendent roots rather than a step from one to the other; both E45 discharges
take their convergence from refinement data and not from the thermody-
namic bound; E78 states that coherence relaxes the supplied cover, and the
coherence-to-content estimate of Eq. (6) bounds disagreement only for en-
coders already shared on overlaps. The inferential content of the composition
is its last step, where contraction and a fixed glued section give uniqueness.
What composing them establishes is that the premises are consistent, named
in one place and machine-checked together, and not that coherence follows
from installed energy or self-representation from coherence.

Joint witnesses combine a register, boundary, refining mesh and small
spatial cover; active witnesses add controlled thermal updates and a priced
installation. They establish satisfiability under their stated identifications.
Mesh convergence follows from refinement data, and the witnesses establish
no common cortical mechanism. Supplemental Material maps the claims to
their Lean declarations and records the counterexamples. The development
declares no additional axioms beyond its libraries' foundational axioms.

705 8 What the observations establish

706 The composition holds under its premises. Whether observation can estab-
707 lish those premises is a separate question, and the three requirements it
708 connects concern different objects: phase dynamics, local descriptions and
709 internal encodings. Their observables must retain these distinctions. The
710 following controls show which mechanisms remain indistinguishable, which
711 temporal approximations fail, which states one observable cannot separate,
712 and what independent content measurements must establish. Each is a finite
713 numerical regime on a declared model, with full parameters and controls in
714 Supplemental Material, and none of them calibrates a cortical quantity.

715 8.1 Stationary distributions leave the mechanism un- 716 determined

717 Equation (2) gives a concentration–coherence relation with no fitted shape
718 parameter. Estimate r from circular resultant length and a from density
719 shape: inverting R from the measured r would make agreement automatic.
720 Even independently estimated agreement identifies a statistical family shared
721 by different mechanisms.

722 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
723 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
724 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
725 oscillators with prescribed drift $-h(t)\sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,

726 and diffusion D_0 have that same density. Their stationary distributions co-
727 incide window by window despite different causes.

728 Resolved dynamics supply additional information. In the ideal model with
729 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
730 increment means constrain drift. Separating endogenous coupling from a
731 matched common drive additionally requires a measured drive or controlled
732 perturbation; identifying electromagnetic coupling requires evidence specific
733 to that interaction. Supplemental Material gives the equivalence construction
734 and its numerical check.

735 **8.2 Finite ramps change the observable onset**

736 The square-root law in Eq. (3) describes stationary solutions. A changing
737 coupling requires time for the phase distribution to approach those solutions.
738 In a deterministic mean-field control that crosses the threshold symmetrically,
739 none of the scanned rates meets the declared tracking tolerance: slowing
740 the crossing does not remove the terminal residual under that initialization.
741 A slow control variable therefore supplies no general guarantee of branch
742 tracking at onset. Supplemental Material gives the residual, rate range and
743 admissible rates away from threshold.

744 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
745 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
746 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay
747 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The

748 6 uncensored legs supply the delay scaling below; the decade-spaced ones
 749 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
 750 onset exponents differ from those of the exact stationary branch evaluated
 751 on the same grids. At the fastest speed the ensemble mean never reaches the
 752 fit window, so its fitted slope does not estimate stationary onset.

753 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
 754 makes the apparent exponent depend on ramp speed and fitting window.
 755 The specified deviation-based critical ramp speed is not bracketed by this
 756 sweep. With fluctuating coupling, the relevant stationary comparison also
 757 depends on rate: the mean coupling approximates order in the tested fast
 758 regime, while averaging stationary order over coupling works for the tested
 759 slow drives confined to one side of threshold. Slow crossing drives can satisfy
 760 neither approximation.

761 **The prediction.** The delay scaling is what this section supports as a pre-
 762 diction, and it needs no absolute calibration. Over the 6 uncensored legs the
 763 mean delay follows $\Delta K \propto v^{0.444}$, with a 95% slope interval of [0.394, 0.494],
 764 which does not contain 1/2. What that shortfall measures is the escape cri-
 765 terion and not the scaling. Applied to a deterministic mean-field run seeded
 766 at the fluctuation floor, the same estimator returns 0.425 over the same legs
 767 at the same escape criterion, inside the interval on the measurement. That
 768 run has no noise to delay it, so the shortfall is the criterion's: it crosses the
 769 saturation between the floor it starts from and the level it is stopped at, and

770 tightening the criterion to $r \geq 0.05$ returns 0.490. The bias is reproduced
771 where the answer is known, which is what licenses reading the measurement
772 as the law rather than as a competing exponent.

773 Stated as a law rather than as a number: onset delay scales as the square
774 root of the rate at which coupling crosses threshold. Testing it requires
775 only that $\dot{K}$ be monotonic in a manipulated rate—not its value, nor the
776 installation factor κ , nor the field-to-coupling conversion rate γ of Eq. (17).
777 It is therefore a within-subject design varying the rate of emergence from
778 anaesthesia alone, each participant serving as their own control. Such a
779 design must avoid the hazard this sweep exhibits: a fixed absolute escape
780 criterion meets a size-dependent critical fluctuation floor, so the criterion
781 has to be referred to each recording’s own pre-threshold fluctuation scale
782 rather than to a shared number. The diffusion D is the remaining input,
783 and Section 8.1 notes that it is estimable from resolved phase traces through
784 the rate $2D$ at which quadratic variation accumulates, which is a different
785 quantity from the concentration the propofol exercise could not range over.

786 Additional controls address integration bias, population size and spatial
787 or signed coupling; the refined signed-network crossing bracket $K_{\text{eff}}/D \in$
788 $[1.958, 2.009]$ contains 2.

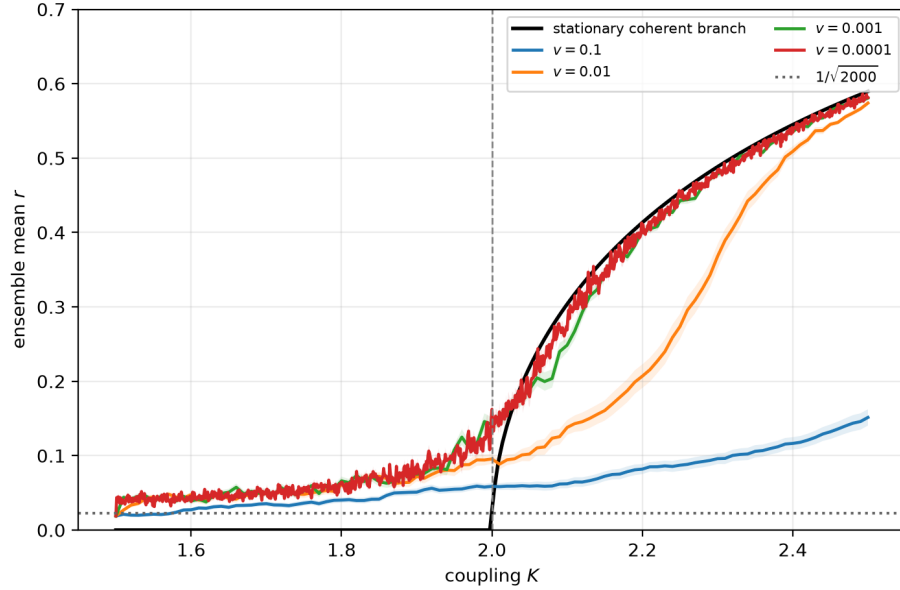


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

789 8.3 A winding state leaves the order parameter am- 790 biguous

791 The order parameter r is indifferent to where a phase sits in space. A field
792 winding once across a periodic sheet therefore has $r = 0$ while every patch
793 of it stays locked. Such a field is not an artificial construction here. On a
794 torus a winding is stationary for any isotropic symmetric kernel at identical
795 frequencies: the coupling term is odd in the separation against a kernel even
796 in it, so the drift cancels pair by pair. `char_is_kuramoto_trajectory` proves

797 this for any finite periodic array, this sheet included. A winding satisfies the
798 positivity and identical-frequency hypotheses. It violates only the quarter-
799 turn initial spread that the convergence result additionally requires before
800 placing the limit at the uniform state.

801 On the sheet of Section 3's spatial control, a winding survives below
802 an interpolated decay length of 0.1871 mm and unwinds above it (Fig. 4).
803 That boundary is interior to the 0.1–0.3 mm band. At 0.1 mm the same
804 sheet at the same coupling, diffusion and kernel reaches $r = 0.9657$ from a
805 uniform start and $r = 0.0083$ from a winding one. Both are stationary, with
806 patch-local order 0.9015 and no winding defect. At fixed K the observable
807 therefore takes two values spanning nearly its whole range. Coupling does
808 not determine the resultant, and a low resultant does not by itself distinguish
809 weak coupling from a winding field.

810 Under the quenched frequency spread of 1.5 rad s^{-1} that the spatial con-
811 trol carries, the boundary moves to 0.1421 mm. The same comparison at
812 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realizations.
813 Heterogeneity shortens the range that sustains a wave without removing it.

814 Which windings persist has a criterion behind it. Linearizing the phase
815 model about a winding returns a coupling of the same translation-invariant
816 form, each separation reweighted by the cosine of the phase the winding
817 advances across it (`hasDerivAt_charJacobian`). That reweighted kernel is
818 again even in the separation, so the same characters diagonalize it and ev-
819 ery mode's rate is one of its eigenvalues. Where the winding advances by

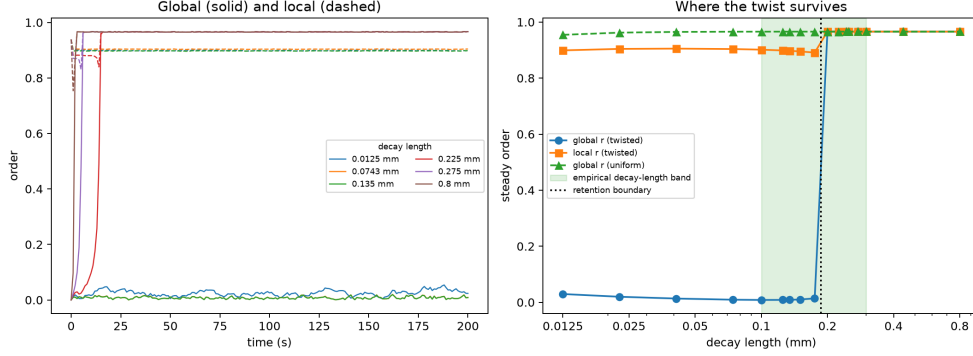


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

820 at most a quarter turn across every separation the coupling reaches, the
821 reweighted kernel is nonnegative and no rate is positive (`charJacobian_`
822 `stable_of_quarter_turn`); where some rate is positive, that mode grows
823 without bound (`charJacobian_growing_mode`). Moving coupling mass to
824 longer separations is what carries a phase advance past a quarter turn, and
825 that is the mechanism the decay-length sweep varies. The criterion bounds
826 rates rather than trajectories: asymptotic stability is not proved, so the
827 states that persist occupy a narrower range than it admits, and the bound-
828 ary quoted above is measured and not derived from it. The same kernel is the
829 potential's curvature, so a winding at $r = 0$ is not the global minimum the

830 convergence result places its limit at, and one carrying a growing mode is not
831 a local minimum either: that result's quarter-turn hypothesis must exclude
832 a winding rather than merely decline to cover it (`char_not_potential_min`).

833 An amplitude layer decides which of these states exist, and leaves what
834 the resultant reports unchanged. Every oscillator of the phase model sits on
835 the unit circle, so a field with an amplitude node has no representation in
836 those dynamics. The real Stuart–Landau extension supplies one on the same
837 coupling, and leaves intact the gradient structure the convergence results rest
838 on. The state becomes $z_i \in \mathbb{C}$ with $\dot{z}_i = (\Gamma - |z_i|^2)z_i + \sum_j A_{ij}(z_j - z_i)$. Its
839 phase equation is the Kuramoto equation with each coupling weighted by a
840 ratio of amplitudes (`amplitude_phase_is_kuramoto`). The phase model is
841 this one with the amplitude ratios set to one, rather than a separate object.

842 What the added degree of freedom decides is which windings exist at all.
843 On a periodic array of any dimension, this sheet included, a winding requires
844 $\Gamma + \lambda > 0$, with Γ the layer's linear gain. Here $\lambda = \sum_d f_d(\cos \Delta\theta_d - 1)$ is
845 the eigenvalue the kernel supplies, summed over separations, with $\Delta\theta_d$ the
846 phase the winding advances across d (`exists_charState_iff`; on the ring
847 $\Delta\theta_d = 2\pi qd/n$). For the nonnegative kernels used here $\lambda \leq 0$, and moving
848 coupling mass to longer separations lowers it further. A broad enough kernel
849 therefore removes the winding outright. The sweep above reaches that same
850 boundary by a different route: the phase model retains every winding at
851 every coupling, and loses this one under integration instead. The two layers
852 bound the same regime from opposite sides. A phase recording does not

853 separate a state the amplitude layer excludes from one the dynamics fails to
854 sustain.

855 Three limits bound the layer for a reader of the cortical literature. These
856 states are windings and not spirals, and on the sheet that is a theorem rather
857 than a shortage of dimensions. Equal modulus at every site puts the phase
858 circulation at exactly zero on every plaquette, and on a periodic sheet the
859 circulations of *any* field sum to zero, so a lone defect is not a state of this
860 geometry at all. Relaxational dynamics has no propagation speed, which is
861 the quantity a cortical wave measurement leads with. And $|z|$ is a normal-
862 form radius, which a measured envelope is not; this work identifies the two
863 nowhere. Supplemental Material gives the sweep, its uniform control, the
864 amplitude layer, the standing wave it admits inside the band, the potential's
865 curvature at a winding and the defect theorems, together with the remaining
866 limits of both.

867 A vanishing resultant is the absence of global agreement and not of agree-
868 ment. Eq. (6) read on a nearest-neighbour patch of a winding gives a discrep-
869 ancy bound of $2\sqrt{2} L |\sin \Delta\theta|$, with $\Delta\theta$ the phase advanced across the patch,
870 which falls to zero as the same winding is spread over more sites (`chord_le_`
871 `of_char_patch`), while the global form of the same bound is vacuous at $r = 0$.
872 Encoders shared on overlaps therefore still bound content disagreement in-
873 side a patch of a state that no global observable calls coherent. The winding
874 number itself stays unconnected to content. It is, however, a property of the
875 state rather than of the cover read on it. On a ring the whole turns dropped

876 by the winding's affine local lifts are zero at every site but the wrap, where
 877 the residue resets, and they sum around the loop to q ; a nonzero q obstructs
 878 every global real-valued phase lift, so one state at one coupling has resultant
 879 exactly zero and degree q (`winding.degree obstructs`). Reweighting the
 880 sites moves the first quantity and redrawing the loop's intermediate transi-
 881 tions leaves the second where it was. The scope is the ring's: this is a loop's
 882 winding and not a spiral, the transitions are those of the affine lifts rather
 883 than of an unwrapped measurement, and the integer is identified with no
 884 cortical observable.

885 This qualifies what a phase trace can establish. An onset criterion stated
 886 in r alone registers a change of basin as readily as a change of coupling.
 887 The protocol of Section 9.3 therefore needs a spatial observable beside the
 888 resultant.

889 **8.4 Coherent adaptation can lose environmental align-** 890 **ment**

891 An adaptive coupling rule might supply representational structure, but its
 892 objective must reward the relevant agreement. For symmetric coupling, pair-
 893 wise sine terms cancel in the population mean drift. Minimizing squared drift
 894 at that fixed mean therefore rewards equalizing drift across nodes. In the
 895 frequency-cluster control, this favors coupling between unlike-frequency clus-
 896 ters, whereas the diagnostic environmental template rewards coupling within

897 clusters.

898 The formalization proves descent of the squared-drift objective for a spec-
899 ified gradient flow at fixed phases. Its functional differs from thermodynamic
900 entropy production: at a coherent stationary density the probability current
901 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
902 rial gives the operator, its gradient and the conditions of the descent result.

903 The finite control co-evolves noisy phases and symmetric, nonnegative
904 couplings at fixed total resource. Its template is a diagnostic covariance
905 of latent signals, never a training input. Across three illustrative seeds,
906 second-half order remains between 0.9167 and 0.9222, while the within-
907 cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–
908 0.6135 (Fig. 5). Random-update, permuted-gradient and frozen-kernel con-
909 trols are reported in Supplemental Material.

910 The immediate post-update objective reduction disappears when aver-
911 aged over complete update intervals. A prespecified grid of 6 rate–cadence
912 combinations contains no case meeting the 5% sustained-reduction criterion
913 while retaining coherence and losing alignment. The finding is coherent adap-
914 tation away from the specified structure, with no demonstrated sustained cost
915 reduction.

916 Kernel alignment and overlap compatibility are different observables.
917 This control addresses acquisition of its designated template under its par-
918 ticular objective. Learning models can explicitly include data in an objective
919 (Corberi et al., 2026); an account of agreement about the cup must specify

920 the content its learning rule is meant to preserve.

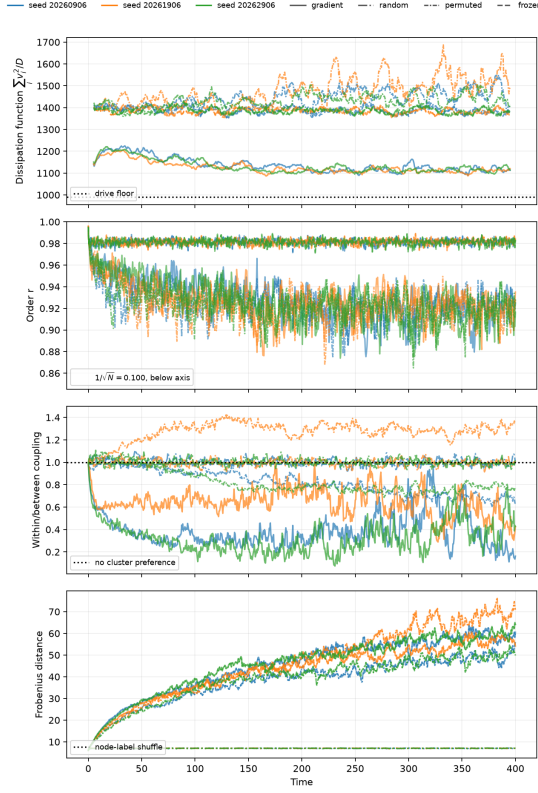


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

921 8.5 Content agreement needs independent decoders

922 A neural test needs independently identified domains, local descriptions and
923 restriction maps. Two decoders should estimate a shared content on over-
924 lapping domains before their agreement is examined. Discrepancy requires a
925 null under decoding noise and controls using deliberately compatible and in-
926 compatible examples. An implementation on synthetic data separates these
927 cases at $\text{AUC} = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial;
928 the compatible construction scores 0.476 on an interval containing one half.
929 Shared-prior shrinkage can nevertheless manufacture agreement, and regional
930 bias can manufacture disagreement. Per-patch decoding diagnostics detect
931 these failures; the compatibility statistic alone does not.

932 Phase agreement cannot replace that measurement. On constructed con-
933 tent, phase carries a share 1.000 of the content entropy when content is built
934 from it and 0.000 when it is not. A phase-derived statistic reports com-
935 patibility on phase-locked patches whose local descriptions disagree. The
936 implemented content statistic detects their incompatibility. These controls
937 validate a measurement on known constructions; a neural content model and
938 decoder remain to be established.

939 Choosing the cover also chooses a candidate subject: trivial descriptions
940 can agree while carrying little content, and a partition can conceal disagree-
941 ment. The useful choice must therefore be constrained by what neural ac-
942 tivity encodes. The exact composition assumes compatibility of its supplied
943 cover. The content mechanisms above identify additional hypotheses under

944 which agreement can arise; they leave the empirical choice of content and
945 the connection to that cover explicit.

946 **8.6 Estimator calibration on exploratory EEG**

947 The concentration range and the dependence structure decide whether the
948 concentration-coherence relation can be tested at all. Under the specified
949 log-density estimator and bias-floor criterion, discrimination at the observed
950 concentration ceiling $a = 0.542$ requires 3000 independent sites in the scanned
951 designs, rising to 31000 when sites share a mean in groups of 10. These
952 counts describe that estimator and dependence model; the intended recording
953 requires its own calibration. Supplemental Material reports the bin-count
954 controls and the full dependence-aware design.

955 That ceiling comes from exercising the estimators on public propofol-
956 sedation EEG (Bajwa et al., 2025), pooling channels in 100-ms win-
957 dows. Bipolar and common-reference montages produce different residuals.
958 Bipolar-montage scalp EEG sits in the linear Bessel regime ($a_{\max} = 0.542$,
959 deviation from $a/2 < 1\%$). The exercise confirms the estimator is well-
960 behaved in this regime but does not test the nonlinear prediction. The
961 bipolar results are compatible with the Bessel relation and with any mono-
962 tone alternative in this regime. Temporal pooling changes the estimand, and
963 sedation recordings leave the awakening hypothesis untested.

964 An informative dataset therefore needs sufficient concentration range or
965 precision, calibrated estimators and an observation model for dependence,

source mixing and reference choice. Testing the nonlinear prediction requires intracranial recordings (ECoG, sEEG or LFP) where local synchrony reaches $a > 1$, or narrowband alpha-spindle burst analysis, at the sample sizes established above ($\geq 31,000$ pooled phase samples or ≥ 1000 independent sites). Public mouse calcium imaging provides thousands of spatially located neurons across waking, sleep and anesthesia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the phase-estimation band used here. Site count, state coverage and bandwidth are separate requirements.

9 The cortical hypothesis and an awakening test

The cortical proposal identifies a physical source of coupling and asks whether its changes help explain recovery. The tests in Section 8 determine what observations must distinguish before a phase trace can support this identification.

9.1 Physical identification and calibration

The cortical hypothesis identifies part of effective coupling with endogenous extracellular electromagnetic interactions. Fields can modulate spike timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models propose coordination of distributed neural information (Pinotsis & Miller,

2022; McFadden, 2020). Phase organization under propofol motivates study-
ing this regime (Bastos et al., 2021), but does not measure coupling.

The formal field predicate requires a jointly continuous kernel, a normal-
ized substrate, coupling equal to the kernel average, and no single site pro-
viding more than half that strength. Toy constructions satisfy it. A cortical
realization must specify the scale at which the kernel and its reduction ap-
ply: calcium imaging reports different state-dependent modularity patterns
at single-cell and mesoscale resolution (Kiyooka et al., 2026).

Cortical columns are the obvious candidate unit, and they answer one
of the two demands the formalism makes. As the oscillator index they are
a serious proposal: they are delineated by response properties and anatomy
rather than by the agreement they would be used to establish (Mountcastle,
1997), and a phase can be estimated from a column-sized territory at the
electrode spacings used for cortical wave measurements (Townsend et al.,
2015). As the content cover they are a poor one, for three separate reasons.
A parcellation into non-overlapping columns is a disjoint cover, which satisfies
the compatibility obligation without constraining anything (Section 4). The
cover condition additionally requires the patches to exhaust the substrate,
while much of association cortex has no accepted columnar parcellation. And
the column’s standing as a canonical cortical unit is contested (Horton &
Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
supplies the oscillator index and leaves the content cover to be identified on
separate evidence, which is the bridge assumption E78 records.

1008 The spatial reduction is also testable. A row-sum aggregation at zero
 1009 diagonal tracks the uniform-coupling control over the tested spatial decay
 1010 range, while frequency heterogeneity displaces the operational threshold. The
 1011 fit window biases both estimates relative to $2D$; agreement with the control
 1012 is therefore the supported conclusion. Supplemental Material reports the
 1013 reduction, its numerical error and the distinct assumptions needed for the
 1014 kernel-average identification in Eq. (5).

1015 The field commitment requires physical evidence beyond a choice of math-
 1016 ematical representation, and the available support comparisons do not supply
 1017 it. A finite matrix spread over positive-mass cells preserves its total cou-
 1018 pling and phase correlation exactly, while point-supported vanishing under
 1019 an atomless measure and a disadvantage from excluding a better-correlated
 1020 pair concern specified support constraints; Section 6 reads the same compar-
 1021 isons against a digital candidate. Neither supplies energy prices.

1022 A spike-time shift alone supplies no coupling rate. The dimensionless
 1023 product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and
 1024 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (17)$$

1025 This rate describes how induced timing shifts accumulate into drift. Nei-
 1026 ther it nor the installation factor κ follows from a heat bound. Testing the
 1027 physical threshold requires independent estimates of the interaction kernel,

1028 its aggregation and diffusion in the same cortical regime.

1029 **9.2 Extracellular geometry and recovery**

1030 The proposed control variable is extracellular geometry. Extracellular vol-
 1031 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
 1032 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
 1033 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
 1034 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
 1035 an effective-medium description predicts that a larger extracellular fraction
 1036 raises conductivity and reduces field amplitude.

1037 The coupling-gated hypothesis consequently predicts that extracellular
 1038 contraction on awakening raises E and, holding the other factors in Eq. (17)
 1039 fixed, raises effective coupling. This argument supplies a direction; the
 1040 geometry–coupling relation, its magnitude and its role in human recovery
 1041 require measurement.

1042 For a smooth crossing at nonzero speed, evaluating the stationary branch
 1043 along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (18)$$

1044 This follows from Eq. (3) at fixed D and concerns the stationary branch
 1045 evaluated at $K(t)$. Reading it as observed recovery requires an indepen-
 1046 dently justified tracking approximation. The crossing controls in Section 8.2

1047 leave that approximation unsupported at onset, even though a separation of
1048 geometry and phase time scales can support it away from threshold. The
1049 awakening hypothesis therefore needs a calibrated dynamical prediction for
1050 an operational onset above the finite-size fluctuation floor. Crossing K_c alone
1051 does not supply a recovery latency.

1052 **9.3 A protocol for the awakening experiment**

1053 The physical test has six requirements:

- 1054 1. **Coupling and diffusion in the same units.** Estimate them from re-
1055 solved phase dynamics, separating heterogeneity and observation noise.
1056 Measure the drive or perturb the system to distinguish endogenous cou-
1057 pling.
- 1058 2. **Geometry calibrated to coupling.** Validate the geometry observ-
1059 able and its relation to effective coupling, with uncertainty. The con-
1060 ductivity argument supplies only the expected direction.
- 1061 3. **Calibrated phase measurements.** Estimate r by resultant length
1062 and a by unweighted regression of log density on centered cosine.
1063 Choose site count and binning from a dependence-aware calibration
1064 over the intended concentration range. The tested designs require
1065 thousands of sites at the observed concentrations; temporal pooling
1066 changes the spatial estimand.

1067 **4. An independent recovery endpoint.** Measure behavioural recovery
1068 separately from phase organization.

1069 **5. A predeclared dynamical onset criterion.** Set a coherence thresh-
1070 old above the finite-size floor and evaluate its latency under the mea-
1071 sured coupling trajectory, including ramp speed or fluctuations. Record
1072 a spatial observable beside the resultant, since a winding field reports
1073 a low resultant at coupling that sustains local order (Section 8.3).

1074 **6. Matched model comparison.** Compare coupling increase, diffusion
1075 change, common drive and alternative recovery models under the same
1076 observation and filtering assumptions.

1077 Failure of the concentration–coherence relation over an informative range
1078 constrains the density model. Failure of the calibrated geometry–coupling
1079 association constrains the physical gate. Better prediction of independent
1080 recovery by a competing mechanism counts against its proposed role. A
1081 favorable outcome still leaves the content of coordinated activity to be ex-
1082 plained.

1083 **10 Discussion**

1084 Two things have been established here, and they are of different kinds. The
1085 constraints of Section 2 hold outright: synchrony does not deliver compatible
1086 content, and where shared encoders let it bound content at all, the bound

1087 that holds uniformly over pairs does so only inside patches too small to
 1088 span the territories at issue, and distance costs either a connected cover or
 1089 a locked configuration with a declared spectral gap. The resultant does not
 1090 determine the state; a unique fixed point is not a reconstruction; a region that
 1091 cannot resolve a family cannot encode it; and agreement about a change takes
 1092 communication. Any account that reads unity off phase coherence has to
 1093 meet them, this one included. The account set out in Section 1 is conditional
 1094 at every step by contrast: each connection in it is a hypothesis supplied
 1095 rather than a result derived, and Table 1 names all eight. Its four non-
 1096 standard ingredients, named in Section 1, are the installed-energy condition,
 1097 the patch-local form of the content bound, the criterion separating a winding
 1098 field from weak coupling, and the machine-checked composition with its joint
 1099 witness. What follows takes the conditions in turn, as commitments the
 1100 proposal incurs, as measurements that would bear on them, and as questions
 1101 the framework leaves open.

1102 **10.1 Unity and a minimal self as interpretive motiva-** 1103 **tion**

1104 We identify the state glued from compatible local descriptions as a formal
 1105 correlate of conscious unity, and its accurate internal reconstruction as a
 1106 correlate of a minimal self. This identification is the framework's explicit
 1107 interpretive motivation: it explains why the mathematics might matter. The

motivations are that different contents belong to one experienced situation and that the situation is experienced from a perspective. The framework’s empirical content, however, is the coupling-gated onset prediction and its spatial signature. The conditional results identify relations a realization would need: sufficient installed resources for the specified phase model, agreement about shared content, and an encoding that supports reconstruction at the required fidelity. Discharging E12–E89 for cortex would make the composition a statement about cortex: that a cortical field above threshold relaxes to a cover whose compatible local descriptions glue, and that the glued state is the unique fixed point of a reconstruction map realized there. It would leave the two identifications above exactly where they are, because neither is among the eight and neither follows from their conjunction; they are the motivation this subsection names, and without an independent measurement for experience, they do not form a testable proposal. Section 10.2 lists the empirical observations that would count against the framework’s mathematical claims.

Compatibility allows different descriptions to contribute different information. It also permits uncertainty and conflicting judgments: apparent cup position and judged cup position can be distinct quantities in a consistent description of their discrepancy. Which quantities are shared must be determined independently. Choosing them solely to secure agreement would deprive the proposal of empirical force.

The minimal-self proposal concerns the perceiver’s position and condition

1131 within the situation, such as the cup’s location relative to a body that can
1132 reach it. The encoding could be implicit and need carry no autobiograph-
1133 ical narrative. Its proposed necessity is philosophical: even an unreflective
1134 episode has a perspective. Gluing by itself requires no self-encoding, and re-
1135 constructability alone supplies no functional role. Altering the encoded hand
1136 or viewpoint while holding the external scene fixed should change estimates
1137 of the perceiver’s relation to the cup as the readout predicts. Whether that
1138 represented perspective is experienced remains a further question.

1139 **10.2 What a test of the full proposal must establish**

1140 Both uniqueness results concern specified objects. They choose neither the
1141 cover nor the encoding, and do not select between candidate subjects. In-
1142 tegrated information theory treats exclusion explicitly (Tononi et al., 2016);
1143 here selection requires independent evidence about neural organization. A
1144 constant readout can contract and fix a predetermined state while discard-
1145 ing its input. The family criterion in Section 5.2 tests the sensitivity that
1146 this construction lacks, provided the relevant states and fidelity are declared
1147 independently.

1148 A test must separately measure coupling relative to diffusion, content
1149 compatibility and self-reconstruction, alongside an independent endpoint for
1150 experience. Meeting all three without a unified episode would challenge suffi-
1151 ciency. A unified episode with incompatible descriptions would challenge the
1152 unity identification; one lacking the specified self-encoding would challenge

1153 the reflexive requirement. Awakening, dreaming and dissociative states offer
1154 possible settings, but absence of report cannot by itself define absence of
1155 experience.

1156 The next physical requirement is a calibrated geometry–coupling relation
1157 and resolved phase dynamics. The installation bound contributes a nec-
1158 essary resource condition where its hardware representation applies; cortical
1159 mode prices and field-mediated coupling require measurement. For the repre-
1160 sentational connection, an informative experiment asks whether a controlled
1161 change in coordination alters independently decoded content agreement, with
1162 controls for sensory information and decoder quality. The constructed-data
1163 estimator establishes feasibility for specified contents. Neural local descrip-
1164 tions, shared overlap encoders and the self-encoding have yet to be identi-
1165 fied. The population-size dependence of the phase-to-content bound limits
1166 its quantitative use in the form Section 4.2 reads off the sheet, and each
1167 way around it asks the measurement for something further: a located excep-
1168 tional set, before a count of disagreeing pairs bears on a named overlap; a
1169 cover whose nerve is connected and whose patch diameters are measured; or
1170 a frequency-locked population over a coupling graph whose connectivity is
1171 measured structurally rather than estimated from the activity the bound is
1172 about.

1173 The separation results explain why these measurements must be inde-
1174 pendent. A stationary distribution leaves several recovery mechanisms indis-
1175 tinguishable. Equal phase order can accompany incompatible descriptions

1176 or different task information. The tested plasticity rule preserves coherence
1177 while losing its designated template. Each finding concerns the specified
1178 model and observable; their role is to delimit the evidence needed for the
1179 proposed connections.

1180 **10.3 Explanatory scope**

1181 Integrated-information, global-workspace and predictive-processing accounts
1182 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
1183 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a
1184 candidate physical interaction and conditional relations between local content
1185 and self-encoding. Gluing and reconstruction permit multiple substrates;
1186 the electromagnetic commitment enters through physical calibration. That
1187 permission is wider than the unconditional results of Section 2 alone. The
1188 conditional chain is substrate-neutral in the same way: its field condition
1189 asks for a jointly continuous kernel on a normalized substrate, a coupling
1190 equal to that kernel’s double average, a noise that is the field’s own, and
1191 no single site carrying more than half the coupling—and none of these dis-
1192 tinguishes an electromagnetic kernel from any other continuous mean-field
1193 kernel. The site condition is the only one with physical flavour, and it holds
1194 vacuously wherever the substrate has no atoms (`no_site_dominates`), which
1195 is the continuum case the electromagnetic hypothesis is about. The chain
1196 therefore transfers to extracellular ion diffusion, to gap junctions modelled
1197 as a density, or to a mechanical or optical medium, modulo the calibration of

1198 Section 9.1. What carries the electromagnetic identification is that calibra-
 1199 tion together with the extracellular-geometry argument of Section 9, which is
 1200 the one place a field equation enters. The formalization checks implications
 1201 under their hypotheses, and its witnesses establish realizability on declared
 1202 models. Two things stand outside that result and are not supplied by it: the
 1203 eight hypotheses are discharged for toy components rather than for cortex,
 1204 and the identification of the glued state with unity and of its reconstruction
 1205 with a minimal self is a further commitment that the eight do not contain.
 1206 The explanatory question is whether the specified relations occur in neural
 1207 activity and matter for experience. In the cup example, this means establish-
 1208 ing how coordination helps maintain one represented situation and how the
 1209 perceiver’s place within it contributes to the episode. The proposal identi-
 1210 fies candidate correlates and tests of that identification. Why those relations
 1211 would be accompanied by experience remains unexplained.

1212 Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i
N	Eq. (1)	Number of oscillators in the population the quantity is read on
D	§2, Supp.	Phase diffusion in inverse time; phase spread $D \leq \pi/2$

Table 2 (continued)

Symbol	Defined	Meaning
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 8.3	Winding number of a phase field read around a ring
μ	Eq. (4)	Substrate measure, normalized to one
Ξ, ϕ_i, u_i, c_i	Eq. (4)	Actuator configuration; mode profile, installation price and occupancy of mode i
U_{inst}, κ	Eq. (5)	Installed energy and the mode-data conversion factor bounding $K_{\text{mf}} \leq \kappa U_{\text{inst}}$
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances
M, b	Eq. (10)	Size of a declared family of states and code length in bits
Λ, τ	Eq. (8)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time
T	§3.2, §5	Temperature in the thermal scale $k_B T$; number of communication rounds to a deadline
ν	Section 6	Fan-in of a communication graph: the most sites any one site hears from
$k_B T, B$	Eq. (16)	Thermal scale, read as one symbol, and heat budget of a register
$\gamma, \mathcal{T}, f$	Eq. (17)	Field-to-coupling conversion rate, timing sensitivity and frequency
α	Section 9	Extracellular volume fraction
λ	§8.3, §9	Eigenvalue the coupling kernel supplies at a winding; extracellular tortuosity
Ω, λ_2	Section 4.2	Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph
β, k	Section 4.2	Chord diameter of a patch and the diameter of a cover's nerve, in hops

Table 2 (continued)

Symbol	Defined	Meaning
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
Φ, R	Eq. (10)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (8)	Encoding region and the reconstruction map $F(s) = \text{readout}(s \mathcal{A})$
$\mathcal{E}$	Eq. (15)	Energy observable shared by the stages of a continuing protocol
z_i, Γ, A_{ij}	Section 8.3	Complex amplitude, linear gain and coupling matrix of the amplitude layer
$\Delta\theta_d$	Section 8.3	Phase a winding advances across separation d
E	Eq. (17)	Field amplitude, in the cortical identification
U	§3.2	Stored energy $U(\Xi)$ of a configuration
s, s_*	§4–5	A global section and the one the reconstruction fixes

Data and code availability

The Lean development, simulation scripts, tests and saved numerical summaries are maintained in the GitHub repository `bobaseb/coupling_gated_cortical_coherence`. Publication simulation values are generated from saved summaries without rerunning production sweeps. The exploratory EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025) (subjects 1010, 1016, 1022, 1033, 1045, 1056, resting awake and sedation runs 1–2); preprocessing and selection are described in Supplemental Material and implemented in `simulations/empirical_collapse.py`. The repository contains the reproducible analysis and does not redistribute the raw record-

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1227 **Use of AI tools**

1228 AI assistance to this work was substantial and is not separable by task. Sev-
1229 eral coding assistants and chat interfaces were used in an interleaved way
1230 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
1231 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
1232 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
1233 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,
1234 accessed with the Hermes CLI. Between them they contributed to the Lean
1235 development, the simulation and analysis code, and the drafting and revision
1236 of both documents. Because the tools were used in combination and in alter-
1237 nation, no section, proof, figure or result is attributable to any one of them,
1238 and none is claimed to be. The author is responsible for all content, including
1239 everything a tool produced. Machine checking establishes the stated Lean
1240 results under their hypotheses; it does not validate biological interpretations
1241 or replace scientific review of generated code and prose.

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